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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination APRIL 2021

MT203S – CALCULUS

Time : 3 hrs **Max Marks :75**

SECTION - A (5X2=10)

Answer **ALL** Questions

1. If $x = r^2 - \theta^2$ where $y = 2r\theta$, find $\frac{\partial r}{\partial x}$.
2. Find ρ at the point $(-2,0)$ of the curve $y^2 = x^3 + 8$.
3. Find the envelope of the family of straight lines $y = mx + \frac{a}{m}$.
4. Show that $\Gamma(n+1) = n!$.
5. Find the centroid of the area enclosed by the parabola $y^2 = 4ax$.

SECTION - B (3X5=15)

Answer any **THREE** Questions

6. If $x = r\sin\theta\cos\varphi, y = r\sin\theta\sin\varphi$ and $z = r\cos\theta$, prove $\frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)} = r^2\sin\theta$.
7. Show that the evolute of the cycloid $x = a(\theta - \sin\theta), y = a(1-\cos\theta)$ is another cycloid.

8. Find the asymptotes $\left(x^2 - y^2\right)y^2 = 4y^3 - 16.$

9. Evaluate $\int_0^1 x^m \left(\log 1/x\right)^n dx.$

10. Find the area of the Cardiod $r = a(1+\cos\theta)$.

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) If u is the function x and y; $x = r\cos\theta, y = r\sin\theta$, show that
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y 2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r}\frac{\partial^2 u}{\partial \theta^2}.$$

(or)

b) Discuss the maxima and minina of $x^3y^2(1-x-y)$.

12. a) Derive the coordinate of the centre of curvature in Cartesian form.

(or)

b) Show that the radius of the curvature $r^n = a^n \cos\ n\theta$ is $\frac{a^n r^{-n+1}}{n+1}$.

13. a) Find the evolute of $xy = c^2$.

(or)

b) Find the asymptotes of $x(x-y)^2 - 3(x^2 - y^2) + 8y = 0.$

14. a) Show that if $u_n = \int_0^{2a} x^n \sqrt{2ax - x^2}\, dx$ then $u_n = \frac{2n+1}{n+2}.au_{n-1}$.

(or)

b) Show that $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

15. a) Find the centroid of the loop of the lemnicate $r^2 = a^2 \cos 2\theta$.

(or)

b) Find the volume and the position of the centre of gravity of the tetrahedron bounded by the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

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